

Day 3: Header Forensics & IOC Extraction Engine Foundation

Key Milestone: Core Email Parsing & Extraction Services

Sprint Focus: Execution of core deliverables for Day 3 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: IOC Extraction Service v1.0, Received Chain Parser, Cockpit Main Navigation Component.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | **Key Modules:** NLP/AI Engine, Explainability, Content Risk scoring

Day 3 Execution: Integrated Named Entity Recognition (NER) pipeline to detect impersonated brand names (e.g., Microsoft, Google, PayPal, SBI, Koyeb).
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | **Key Modules:** Header Forensics, Mail Path, Infrastructure Risk

Day 3 Execution: Developed multi-hop Received header parser to extract client origin IPs, MTA relay timestamps, and calculate transmission transit latency.
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | **Key Modules:** IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 3 Execution: Built async DNS resolver module (aiodns/dnspython) to perform A, MX, NS, and TXT record resolution for sender domains.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | **Key Modules:** Graph Engine, Campaign Detection, Similar Attack Detection

Day 3 Execution: Engineered graph node models for Threat Actors, Target Mailboxes, Malicious IPs, Phishing Domains, and Shared Infrastructure.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | **Key Modules:** Risk Scoring, Case Management, Response Recommendations

Day 3 Execution: Created Case creation service with automated UUID/Case-ID generation (`TT-2026-XXXX`) and initial risk weighting algorithm.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | **Key Modules:** Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 3 Execution: Built the Cockpit Dashboard navigation shell, status indicator badges, and dynamic Case Selection Sidebar.

System Integration & Quality Verification

Cross-Module Integration: End-to-end extraction test: parsing sample phishing .EML files outputting structured IOC JSON payloads.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 3 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 3 MILESTONE SIGNED OFF